

# Improved deep residual shrinkage network for a multi-cylinder heavy-duty engine fault detection with single channel surface vibration

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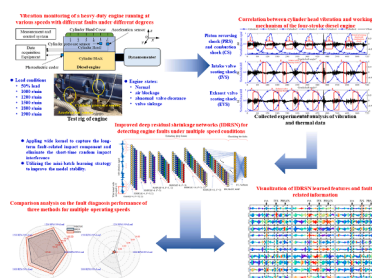
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## HIGHLIGHTS

- An IDRSN method is proposed to diagnose multi-faults at various faulty degrees of diesel engine under different operating conditions.
- The proposed diagnosis approach is validated experimentally using a diesel engine rig test by re-generating the corresponding fault at different degrees.
- Visualized analysis shows good consistency between the learned feature and the fault-related information.
- The proposed method shows high recognition rate, high stability, good anti-interference comparing with other popular algorithms.

## GRAPHICAL ABSTRACT



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## ABSTRACT

The health monitoring and fault diagnosis of heavy-duty engines are increasingly important for energy storage ecosystem. During operation, vibration characters corresponding to the specific fault need to be extracted from the overall system vibration. Faulty characteristics emanating from one single cylinder are also mixed with those from other cylinders. Besides, the change of working condition brings strong nonlinearities in surface vibration. To solve these problems, an improved deep residual shrinkage network (IDRSN) is developed for detecting diverse engine faults at various degrees using single channel surface vibration signal. Within IDRSN, a wide convolution kernel is utilized in first convolution layer to capture the long-term fault-related impacts and eliminate the short-time random impact. The residual network module is adopted to enhance the focus the relevant components of vibration signals. Mini-batch training strategy is used to improve the model stability. Meanwhile, Gradient-weighted class activation map is adopted to assess the consistency between the learned knowledge and the fault-related information. The IDRSN is implemented to diagnosing a diesel engine under various faults, faulty degrees and operating speeds. Comparisons with existing models are analyzed in terms of hyper-parameters, training samples, noise resistance, and visualization. Results demonstrate the proposed IDRSN's superior performance on fault diagnosis accuracy, stability, anti-noise performance, and anti-interference performance. An average accuracy rate of 98.38 % was achieved by the proposed IDRSN, in comparison to 96.64 % and 93.56 % achieved by the DRSN and the wide-kernel deep convolutional neural network

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respectively. These results highlight the proposed IDRSN's superiority in diagnosing multiple faults under various working conditions, offering a low-cost, highly effective, and applicable approach for complex fault diagnosis tasks.

## 1. Introduction

Engines have been widely used for energy storage and conversion, backup power generation, grid stability, and mobility and transportation [1]. To make sure the power plant works in an efficient and stable state, vibration signals are commonly collected for intelligent state monitoring and fault detection [2]. According to thermal and mechanical dynamics characteristics [3,4], the vibration measured from the engine block or cylinder head contains more rich information from cyclic mechanical movements and in-cylinder explosion-induced impact, compared with other signal sources, such as acoustic and thermal signals. A pivotal challenge lies in effectively extracting pertinent fault-related information from the strong interference of noise signals [5].

With the rapid development of artificial intelligent technologies (AI) [6,7], two types of data-driven methods, the based-machine learning methods [8,9] and the based-deep learning methods [10,11], are iterated quickly in the fault diagnosis field. The former is composed of two separate procedures: the feature extraction from collected signals and the pattern classification according to those features. During feature extraction, numerous parameters are calculated to represent specific fault patterns using the complicated vibration signal emitted by a diesel engine. Subsequently, the extracted features are sent into a classifier conducted by the machine learning algorithms to recognize the fault patterns associated with the diesel engine. However, vibration signals measured from a working engine are usually accompanied by strong noise, which diminishes the fault identification accuracy. To eliminate the adverse effect of noise, advanced signal processing methods, such as the empirical mode decomposition (EMD), variational mode decomposition (VMD), and their variants [12–14] are extensively applied to enhance the robustness of fault diagnosis model in the presence of intense noise interference.

Kuai et al. [15] proposed a valve clearance fault diagnosis method based on the wavelet packet decomposition (WPD) and neural networks (NN), in which the WPD is applied to extract the multi-band spectrum power eigenvalues and the NN is treated as a classifier. Since the frequency band features change with the valve clearance, Zhao et al. [14] developed the VMD-random forest (RF) method to recognize the valve clearance fault of diesel engine using the multiband features. To utilize information of multi-channel signals, the multivariate EMD (MEMD) and the dispersion entropy are combined with the support vector machine (SVM) method to diagnose the misfire fault of a diesel engine [13]. In reference [16], a frequency feature extracting method combining FFT and 'DB4' wavelet is introduced to provide features for the SVM to detect the fuel injection fault. The mode aliasing and signal distortion are found in the recursive VMD. Tang et al. [12] improved the signal decomposition approach and employed the component energy distribution to identify the diesel engine faults. In the above research, the pre-processing of signal shows superior performance on feature extraction, but it also increases the time consumption especially when the signals are complicated. Apart from the signal pre-processing methods, more and more vibration features are dug out by the complexity-measuring method [17] (i.e. sample entropy, fuzzy entropy, diversity entropy, multiscale sample entropy, multiscale fuzzy entropy, and their variants) for better pattern recognition. For instance, Ref. [18] develops a diesel engine fault diagnosis approach based on refined composite multiscale fuzzy entropy and the parameter-adaptive support vector machine based on the firefly algorithm. To improve the sensitivity of DE-based method to the signal complexity changes, Ke et al. [19] proposed the multiscale bidirectional diversity entropy and applied

the least-squares support vector machine to the diesel fuel injectors fault diagnosis. The multiscale fluctuation dispersion entropy ignores the high-frequency component information, so the hierarchical fluctuation dispersion entropy is proposed [20]. As we can see the machine learning methods have proven effective in diesel engine fault diagnosis [21], however, it also implies two notable shortcomings in real applications. Firstly, the signal pre-processing and feature-extracting procedure inevitably increases the time required for fault diagnosis and the feature consistency is hard to maintain due to the reliance on self-defined parameters. Secondly, the generalization ability of classifiers tends to be insufficient across diverse cases, which implies that diagnostic performance heavily depends on the mutual selection according to the specific tasks.

Instead, the superposition of convolutional layer within a deep learning-based model is prone to extracting hidden and superior fault-related information and being trained synchronously for feature representation and mode classification [22,23]. In reference [24], the convolutional neural network (CNN) shows an effective diagnosis of abnormal valve clearance of diesel engines. However, the CNN fails to ensure the diagnosis stability when the operating and environmental conditions change. Zhao et al. [25] proposed a multi-branch convolutional neural network with four auxiliary classifiers for diesel engine fault diagnosis. Wang et al. [26] proposed a random CNN structure to utilize ensemble learning for diagnosing diesel engine faults. Qin et al. [27] developed a novel multiscale CNN model with long short-term memory neural network and residual block for misfire fault diagnosis. The receptive field of the CNN model with a small kernel can't capture the internal correlation of vibration signals [28], thus a series of attention mechanism-based models are proposed. Focusing on mutual fault information between different channels, a channel interaction feature encoder based on multi-kernel efficient channel attention was developed for multiple diagnostic tasks of rolling bearing [29]. Vibration signals measured from rotation machines commonly include noise components, worsening the models' feature learning ability. Based on this, the deep residual shrinkage networks (DRSN) based on learnable soft thresholding were developed for the fault diagnosis of rolling bearing and exhibit better anti-noise performance than common CNNs [30]. Compared with the diagnosis of key components such as rolling bearings and gearbox [31], the engine as a reciprocating power source is a complete operating system and includes thermal and mechanical dynamics, etc., performing more complicated noise sources and working conditions. Thus, the low signal-to-noise ratio of engine vibration remains a challenge for the existing diagnosis model.

With the development of deep learning methods, three issues emerge when they are implemented for complicated engine fault diagnosis, given their strong noise-interference working conditions. Firstly, the kernel number of the first convolution kernel layer is crucial since a narrow perception vision may render the model susceptible to random impact disturbances. Secondly, an existing risk of overtraining in each batch potentially leads the model towards local but global optimum with worse stability. Thirdly, there is a demand to enhance the interpretability of deep learning models.

In this paper, an improved deep residual shrinkage network (IDRSN) is developed for diesel engine fault diagnosis. In IDRSN framework, a large kernel size is applied to the first convolution layer, small batch sizes are adopted in each training batch, and the Gradient-weighted class activation map (Grad-CAM) [32] is employed to interpret the model results. The contributions of this paper are summarized as follows:

- (1) An IDRSN method is proposed to diagnose multi-faults at various faulty degrees of diesel engine under different operating conditions.
- (2) The proposed diagnosis approach is validated experimentally using a diesel engine rig test by re-generating the corresponding fault at different degrees.
- (3) Visualized analysis on the working procedure of a diesel engine is conducted to interpret the results of the proposed method.
- (4) The superior performances of the proposed method (i.e.: higher and more stable fault recognition rate, anti-noise, anti-interference performance) have been validated by the comparative and experimental analysis.

The remainder of this paper is organized as follows: Section 2 provides the based IDRSN schematic in detail, Section 3 describes the diesel engine rig test and the vibration response characteristic analysis, Section 4 illustrates the experimental validation of the proposed method based on IDRSN, and Section 5 summarizes the conclusions.

## 2. Improved deep residual shrinkage networks

The deep residual shrinkage network (DRSN) is developed with the residual shrinkage building unit (RSBU) based on wavelet soft thresholding de-noise [30]. The soft thresholding function is expressed as follows:

$$y = \begin{cases} x - \tau, & x > \tau \\ 0, & -\tau \leq x \leq \tau \\ x + \tau, & x < -\tau \end{cases} \quad (1)$$

where  $x$  and  $y$  are the input and output features, and  $\tau$  is a threshold for noise elimination which is determined during the model training.

RSBU is shown in Fig. 1(a), where  $1 \times L \times C$  is the number of samples, the length of the feature maps, and the channel number of the feature maps, and  $K$  and  $F$  denote the number and size of the kernel, respectively.  $X$  is the feature map reduced to one-dimensional vectors by the absolute and GAP operations. After two FC layers, the output is transformed to 0~1 by the sigmoid activation function. Then the thresholds are computed as:

$$\theta_c = \alpha_c \cdot \text{average}_{ij} |X_{i,j,c}| \quad (2)$$

where  $\theta_c$  and  $\alpha_c$  are the thresholds of the  $c$  th channel of the feature map

and the output of the second FC layers, and  $i, j$ , and  $c$  are the number of feature maps, length of the feature maps, and channel number of the feature maps, respectively. Using RSBU, the useful channels of the feature map are highly weighted, and the irrelevant channels are given small weightings.

As shown in Fig. 1 (b), the DRSN is composed of a convolutional layer with a small kernel size  $3 \times 1$ , several denoised blocks based on RSBU, and a classified module with BN, ReLU, GAP, FC, and *softmax* unit. ‘/2’ stands for the input feature maps, and the output of the forward layer is processed by the moving average with a stride equal to 2. In the model training process, the channel thresholds for the feature map are optimized with an increase in the number of iterations.

Based on the DRSN, two improvements of IDRSN are implemented, as illustrated in Fig. 2. Firstly, a wide convolution kernel in the first convolution layer is applied to eliminate the influence of random impact and noise existing in the complicated vibration signal of the diesel engine, and the small kernel size is utilized in the following RSBU modules to extract the deep features and further suppress the irrelevant components. Secondly, during the model training, the same optimizer is employed as in [30], and a small min-batch, equaling 16, is adopted to prevent the model from converging a local optimum and enhance the stability of model training.

Meanwhile, the remaining hyperparameters (i.e., learning rate, coefficient of momentum, and penalty coefficients) maintain consistency with those outlined in [30]. Specifically, the learning rate is initialized at 0.1 during the early stages of model training, undergoes a gradual reduction to one-tenth of its original values at the 40th and 80th epochs, and eventually halts at the 100th epoch. This strategy adjustment in learning rate ensures the swift training of the IDRSN model at a higher learning rate initially, followed by meticulous fine-tuning at a lower learning rate.

To illustrate the superiority of the proposed IDRSN model in the subsequent experimental analysis, two alternative models, namely deep residual shrinkage network (DRSN) and wide-kernel deep convolutional neural network (WDCNN), are compared. The model parameters of each are meticulously detailed in Table 1, where ‘CBU’ indicates the combination of the convolution layer and batch normalization. ‘Outsize’ means the output data size of each module within the above three models. Meanwhile, both the DRSN and WDCNN models were trained with the same training hyper-parameters. All models were constructed and trained using Tensorflow 2.7.0.

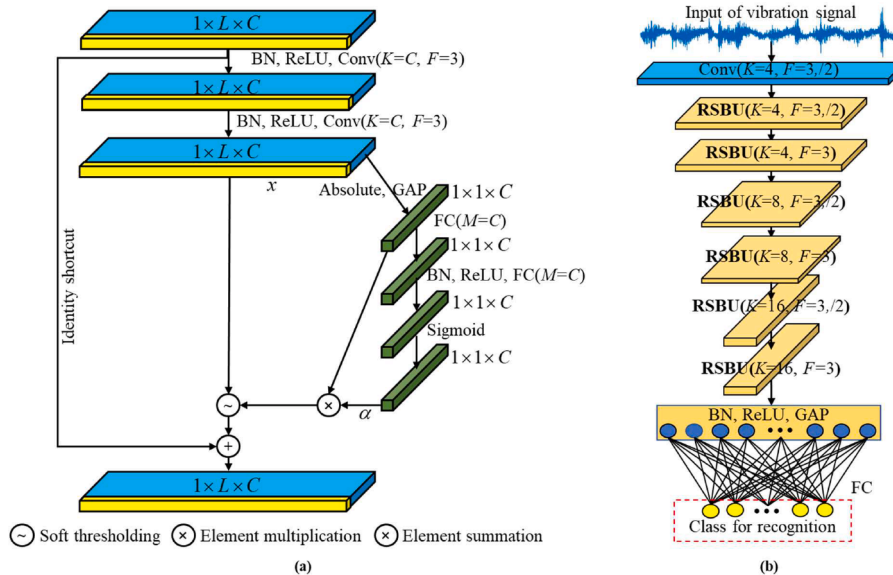


Fig. 1. Schematics of (a) residual shrinkage building unit (RSBU) and (b) DRSN.

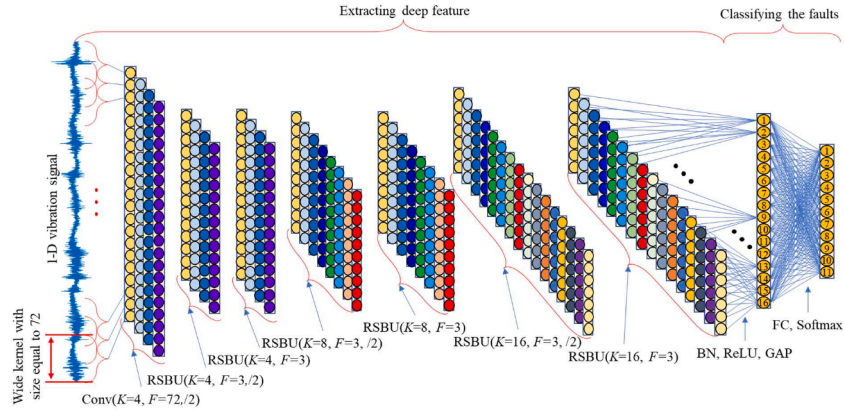


Fig. 2. Improved deep residual shrinkage networks (IDRSN).

Table 1

Model parameters.

| Module | Output size | IDRSN            | DRSN             | WDCNN            |
|--------|-------------|------------------|------------------|------------------|
| 1      | 13,512,031  | Input            | Input            | Input            |
| 2      | 43,256,031  | Conv (4, 72, /2) | Conv (4, 3, /2)  | Conv (4, 72, /2) |
| 3      | 43,128,031  | RSBU (4, 3, /2)  | RSBU (4, 3, /2)  | CBU (4, 3, /2)   |
| 4      | 43,128,031  | RSBU (4, 3, 2)   | RSBU (4, 3, 2)   | CBU (4, 3, 2)    |
| 5      | 8,364,031   | RSBU (8, 3, /2)  | RSBU (8, 3, /2)  | CBU (8, 3, /2)   |
| 6      | 8,364,031   | RSBU (8, 3, 2)   | RSBU (8, 3, 2)   | CBU (8, 3, 2)    |
| 7      | 16,332,031  | RSBU (16, 3, /2) | RSBU (16, 3, /2) | CBU (16, 3, /2)  |
| 8      | 16,332,031  | RSBU (16, 3, 2)  | RSBU (16, 3, 2)  | CBU (16, 3, 2)   |
| 9      | 16          | BN, ReLU, GAP    | BN, ReLU, GAP    | BN, ReLU, GAP    |
| 10     | 11          | FC, Softmax      | FC, Softmax      | FC, Softmax      |

### 3. Data collection

The test rig of a heavy-duty diesel engine, depicted in Fig. 3, includes the diesel engine, the dynamometer, the control system, and data acquisition components. A single-axis vibration accelerometer (1A108E, Donghua, China) is affixed to the cylinder head in the Y direction, close to the sixth cylinder on the exhausting side. Additionally, a cylinder pressure sensor (6125C, Kistler, Switzerland) is plugged into the fifth

cylinder. The data is sampled at a rate of 51,200 Hz, with a duration of 15 s. As outlined in Table 2, three sorts of faults are re-generated in the engine rig test including the air blockage (AB), the abnormal valve clearance (AV), and the valve sinkage (VS). For the AB, four degrees of blockage at 20 %, 40 %, 50 %, and 60 % are induced, which indicates the air intake is blocked at low, medium-low, medium, and medium-high degrees. For the AV, three degrees of “clearance too small” are considered, which are 0.05 mm smaller (low), 0.14 mm smaller (medium-high) and 0.20 mm smaller (high). For the VS, three degrees of sinkage are made, including 0.05 mm sinkage (low), 0.13 mm (medium-high), and 0.20 mm (high). In total, eleven states of engine tests under eleven states are performed, including the normal state (OK), four AB states, three AV states, and three VS states. In particular, the economic speed range is 1000–1400 r/min with a rated speed of 1900 r/min and a maximum torque speed of 1200–1300 r/min. Considering a variety of engine application scenarios, such as heavy trucks and generator sets, the engine tests were conducted under multiple speed conditions, 1000/ 1200 /1500 /1800/ 1900 r/min respectively.

The 15-second data from each operational state is segmented into 700 samples by a sliding window approach, with a window width of 5120 and a step size of 500 points, as depicted in Fig. 4. Considering that the angle-domain data can be more consistent with the engine working

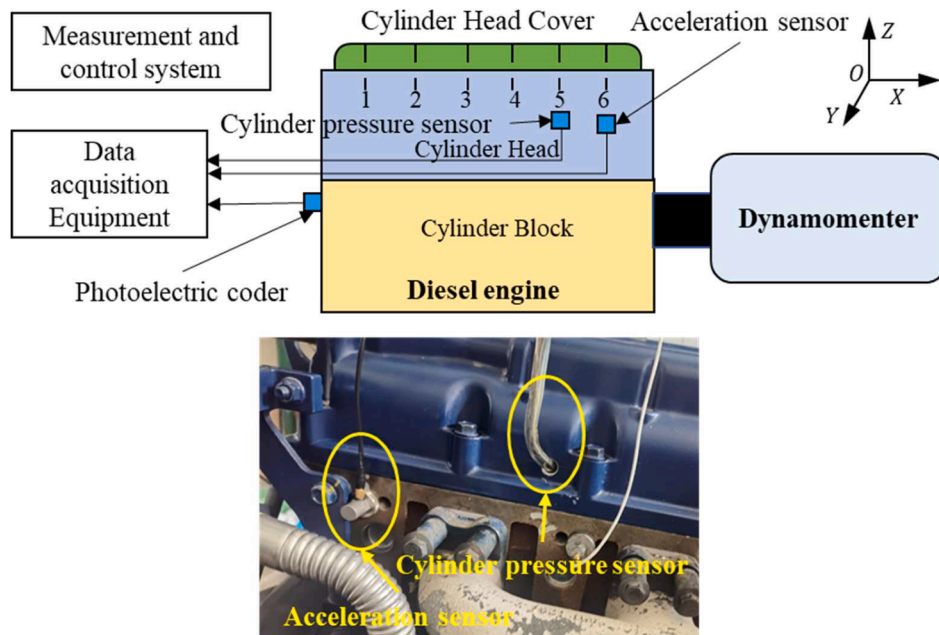


Fig. 3. Diesel engine test rig.

**Table 2**  
Diesel engine states.

| Operating conditions                              | Fault mode                    | Fault degree      | Label | Samples |
|---------------------------------------------------|-------------------------------|-------------------|-------|---------|
| 50 % load @ 1000, 1200, 1500, 1800 and 1900 r/min | Normal                        | —                 | OK    | 700     |
|                                                   |                               | Air blockage (AB) |       |         |
|                                                   |                               | 20 % blocked      | AB20  | 700     |
|                                                   |                               | 40 % blocked      | AB40  | 700     |
|                                                   |                               | 50 % blocked      | AB50  | 700     |
|                                                   |                               | 60 % blocked      | AB60  | 700     |
|                                                   | Abnormal Valve Clearance (AV) | 0.20 mm smaller   | AV20  | 700     |
|                                                   |                               | 0.14 mm smaller   | AV14  | 700     |
|                                                   |                               | 0.05 mm smaller   | AV05  | 700     |
|                                                   |                               | 0.05 mm sinkage   | VS05  | 700     |
|                                                   | Valve sinkage (VS)            | 0.13 mm sinkage   | VS13  | 700     |
|                                                   |                               | 0.20 mm sinkage   | VS20  | 700     |

process, the collected vibration signal data is subsequently analyzed in the angular domain [33].

In Fig. 5, the vibration data from the sixth cylinder head, in-cylinder pressure, and the lifts of the intake and the exhaust valves are depicted for a single working cycle including two revolutions (720 °CA) of the crankshaft. For comparability, the vibration data is normalized within  $[-1, 1]$ , while the in-cylinder pressure and the valve lifts are normalized within  $[0, 1]$ . Consequently, the consistency of the vibration and thermal characteristics was concurrently investigated [33–36].

Fig. 5(a) illustrates the coherence between cylinder pressure and vibration signal during multiple cylinder combustion. When one cylinder of multiple cylinder-engine works, the cylinder pressure sharply increases, and the vibration signal simultaneously reflects the piston reversing shock (PRS) and combustion shock (CS) at the same time. As shown in Fig. 5(a), notable PRS and CS around 600–640 crankshaft angle are present in the vibration signal from the sixth cylinder. Fig. 5 (b) expresses the consistency of the inlet valve shift cure and vibration response during the working of multiple cylinder engine; the inlet valve shift represents the intake working cycle of an engine, and when the inlet valve is from opening to closing, the inlet valve seating shock (IVS) around 460 crankshaft angle is illustrated in the vibration signal from the engine head. Fig. 5 (c) highlights the consistency of the exhaust valve shift cure and vibration response during the working of multiple cylinder engine; the inlet valve shift represents the exhaust working cycle of one engine, and when the exhaust valve is from opening to closing, the

vibration signal from the engine head also illustrates the exhaust valve seating shock (EVS) around 280 °CA.

In summary, the cylinder head vibration response signal accurately reflects the engine thermal process, capturing key events such as PRS, CS, IVS, and EVS. However, owing to the inherent characteristics of a multi-cylinder engine, the cylinder head vibration signal represents the fusion of multiple-cylinder thermal processes. Consequently, the vibration response of one cylinder of a multi-cylinder engine is affected by the responses of other cylinder working processes, which poses a challenge in the engine classification task.

Taking the 50 % load at 1200 r/min as an example, illustrated in Fig. 6, the IVS, compared with the normal state, delays with the decreasing clearance and advances with the increasing clearance. Simultaneously, the vibration amplitudes around the IVS, EVS, PRS, and CS also change due to the variation in clearance. Notably, for air blockage states, although the vibration amplitudes around the IVS, EVS, PRS, and CS differ, the timing of these events remains consistent.

## 4. Results and discussions

### 4.1. Recognition rate

The dataset is randomly partitioned into three segments for each working condition: 400 samples for training, 100 samples for validation, and 100 samples for testing. In particular, the training set and verification set are respectively utilized to update the model parameters and determine the optimal model parameters in the model training process. Meanwhile, the testing sets are used to assess the performance of trained models. The WDCNN, the DRSN, and the proposed IDRSN are implemented to the same datasets for comparison analysis, as depicted in Fig. 7 and Fig. 8.

The accuracy and loss curves of the three methods for both the training and validation sets under 1200 r/min-50 % Load are presented in Fig. 7.

In Fig. 7 (a), the loss and accuracy curves for the training sets of all three methods exhibit noticeable fluctuation in the early 40 epochs and then tend to be constant. However, as shown in Fig. 7 (b), the loss and accuracy cures for validating sets of all three methods exhibit huge fluctuation in the early 40 epochs, and the loss and accuracy cures of the proposed method based on IDRSN tend to be constant, which indicates the superior performance of the IDRSN-based method, compared to the methods based on DRSN and WDCNN with pronounced fluctuation.

Fig. 8 provides insights into the testing classification results of the methods based on the IDRSN, DRSN, and WDCNN under the multiple speed conditions. Notably, for a diesel engine, higher speed corresponds to lower torque under the same 50 % load. Fig. 8(a) displays the classification results of the proposed IDRSN method under multiple speed conditions, and the classification rate of multiple speed conditions is

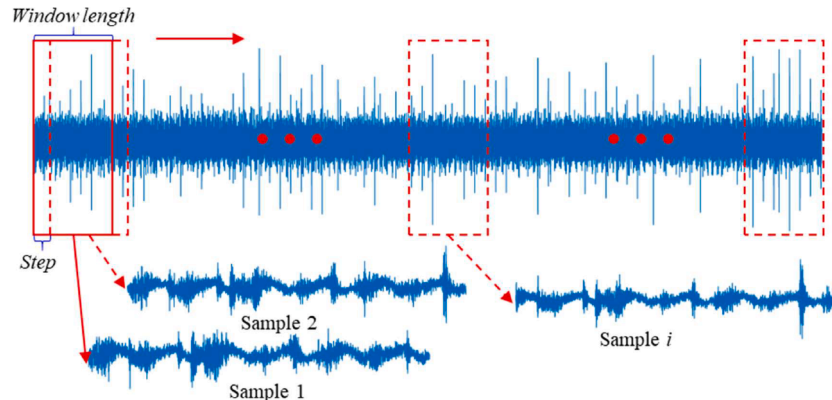
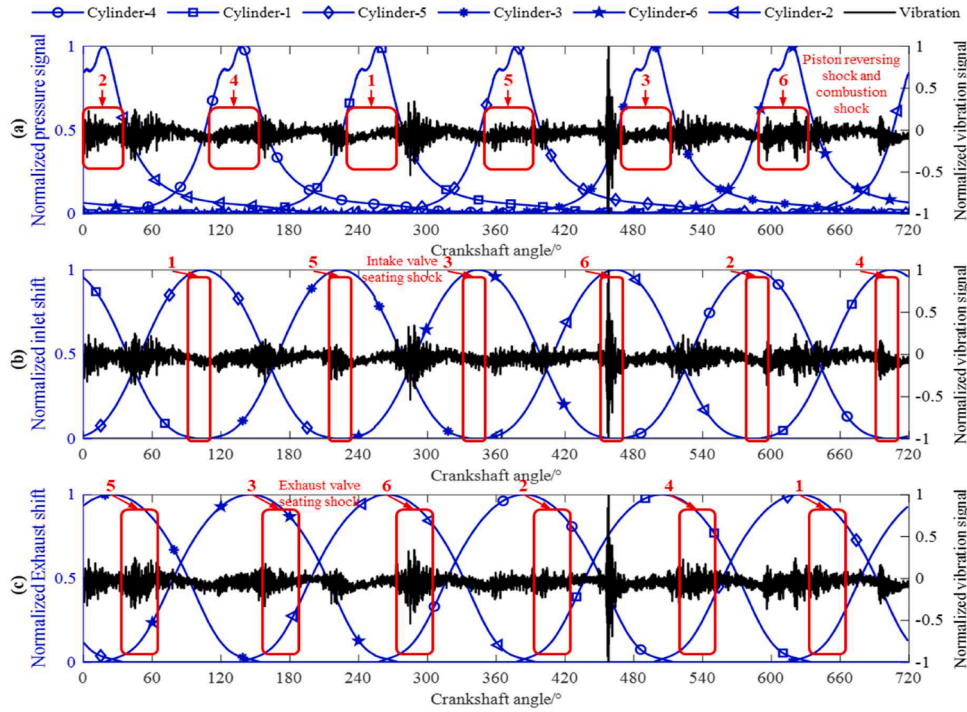
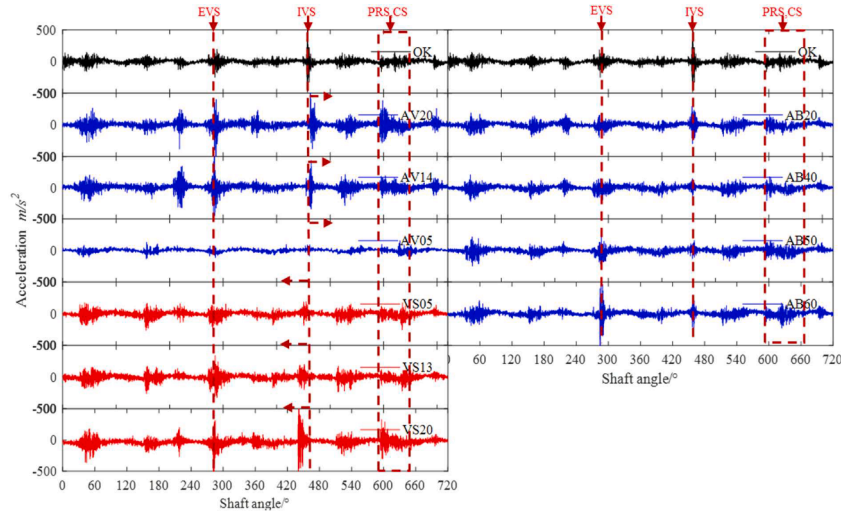


Fig. 4. Data segmentation by sliding window.



**Fig. 5.** Correlation between cylinder head vibration and in-cylinder pressures, intake lifts and exhaust lifts of diesel engine within a single working cycle under normal state.



**Fig. 6.** Angle-domain vibration signal of diesel engine under normal and faulty states.

around 98 %. In contrast, the classification rates of the DRSN-based and WDCNN-based methods fluctuate noticeably. As the speed increases gradually, their classification rates decrease gradually with more misclassified samples. Therefore, Fig. 8 illustrates the superior classification performance of the proposed IDRSN methods to the DRSN-based and WDCNN-based methods, even at multiple speed conditions.

The above results highlight that the proposed IDRSN method, compared with the two methods based on DRSN and WDCNN, outperforms in terms of recognition performance and stability in the validating sets with faster convergence.

#### 4.2. Parameter analysis

To assess the influence of self-defined parameters on the diagnostic performance, the IDRSN model was trained and tested with varying

batch sizes and kernel sizes in the first convolution layer. The experimental data from diesel engine at 1200 r/min-50 % Load were utilized for the analysis. Recognition results, including the average and standard deviation of ten times recognition rates, under different parameters, are presented in Fig. 9 and Table 3. The batch size ranged from 16 to 128 at intervals of 16 and the kernel size varied between 8 and 128 at intervals of 8.

Firstly, with a determined batch size, an increase in kernel size resulted in varying recognition rate averages for IDRSN, displaying larger fluctuations. Conversely, a smaller batch size correlated with reduced fluctuation. Secondly, with the determined kernel size, a decrease in batch size generally led to an increase in the recognition rate averages of IDRSN, exhibiting slight variation and random fluctuations when the kernel size was smaller than 32. Lastly, under larger kernel sizes and smaller batch sizes, the IDRSN-based fault diagnosis method

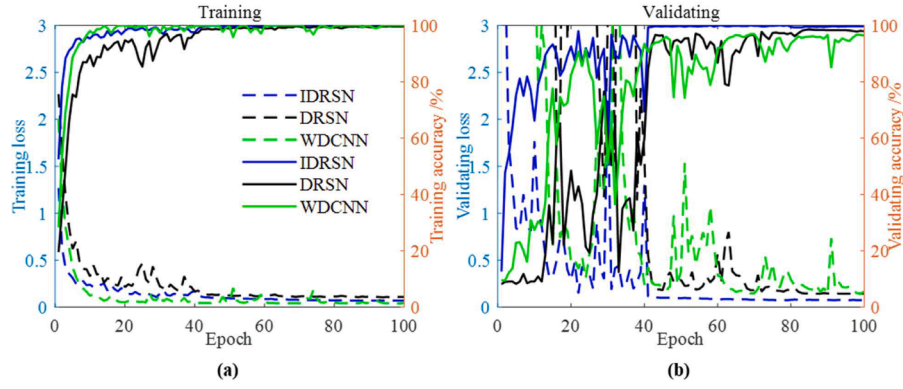


Fig. 7. Losses and accuracies of different methods under 1200 r/min-50 % Load: (a) Training sets and (b) Validating sets.

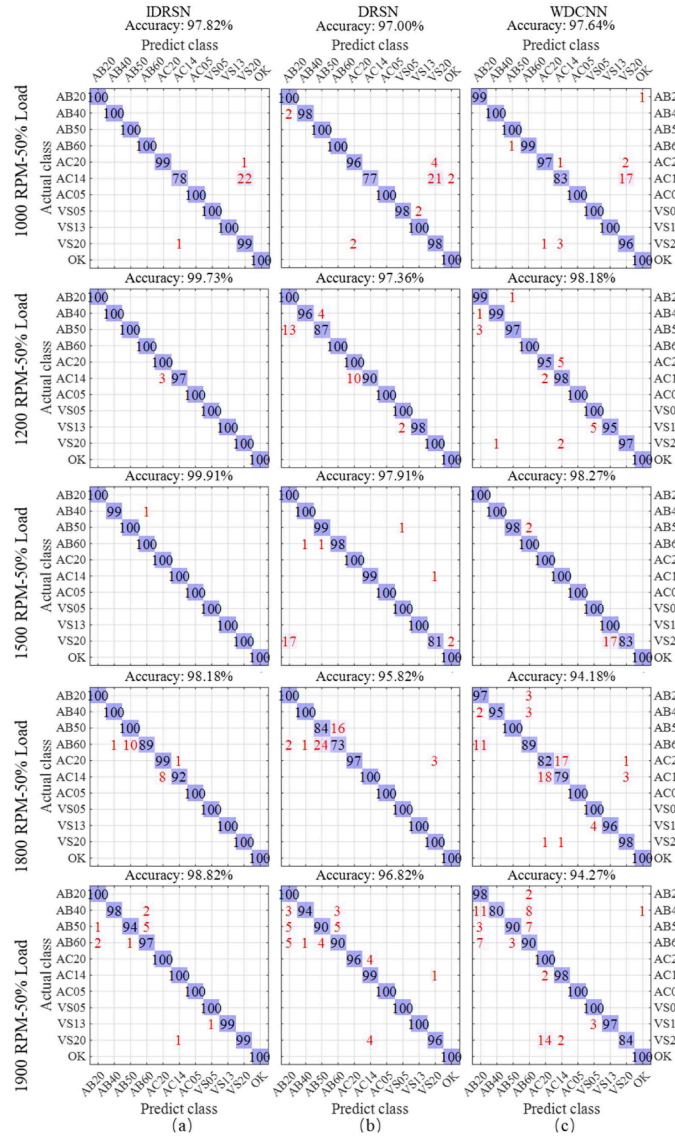


Fig. 8. Diagnosis results of (a) IDRSN, (b) DRSN, and (c) WDCNN.

could achieve better performance in terms of recognition rate with smaller variation. Given the above analysis, the kernel and batch sizes of the proposed IDRSN method were set to 72 and 16, respectively, for subsequent comparative analysis.

#### 4.3. Effect of training samples

To highlight the superior performance of the proposed IDRSN method, a comparative analysis is conducted among IDRSN, DRSN, and WDCNN methods, considering different numbers of training samples and the same number of testing samples. The comparison results under 1200 r/min- 50 % Load condition are presented in Fig. 10 and Table 4. The number of training samples varied in intervals of 50 between 50 and 500 for each state of the diesel engine.

Firstly, as the number of training samples increased, the average recognition accuracy rates of all the methods based on IDRSN, DRSN, and WDCNN primarily increased. However, the recognition accuracy rate averages of the first two methods finally converged toward 100 % with slight fluctuation, while the recognition accuracy rate averages of the third method randomly fluctuated more significantly. Secondly, under the determined number of training samples, the average recognition accuracy rate of the IDRSN-based method surpassed that of the DRSN-based method, which exceeded the average recognition accuracy rate of the WDCNN-based method. Even with a smaller number of training samples, the proposed method retains the superior average recognition accuracy rate. Thirdly, as the number of training samples increased, the standard deviations of recognition accuracy rate for the IDRSN-based and DRSN-based methods decreased, while that of the WDCNN-based methods exhibited larger fluctuation. Generally, the standard deviations of recognition accuracy rate for the IDRSN-based method were smaller than those of the WDCNN-based method.

In summary, compared with the DRSN-based method and the WDCNN-based method, the proposed IDRSN method exhibits the best performance in terms of the recognition accuracy rate and the reliability of model training and testing under the variation in the number of training samples.

#### 4.4. Anti-noise performance

The data at 1200 r/min-50 % Load were taken as an example to demonstrate the anti-noise performance preponderance of the proposed IDRSN-based method, compared with the DRSN-based method and the WDCNN-based method. Comparison results are illustrated in Table 5 and Fig. 11, in which the recognition results of all three methods, under the same vibration signal of the diesel engine with different signal-noise-ratios (SNR), varying from  $-5$  to  $5$ . The smaller the SNR, the stronger the noise component. For a given signal power  $P_s$  and noise signal power  $P_n$ , the SNR can be expressed as [27]:

$$\text{SNR} = 10 \log \frac{P_s}{P_n} \text{ dB} \quad (3)$$

As depicted in Fig. 11 and Table 5, firstly, with an increase in the SNR, the accuracy rate averages of the three methods (i.e., the proposed method based on IDRSN, the DRSN-based method, and the WDCNN-

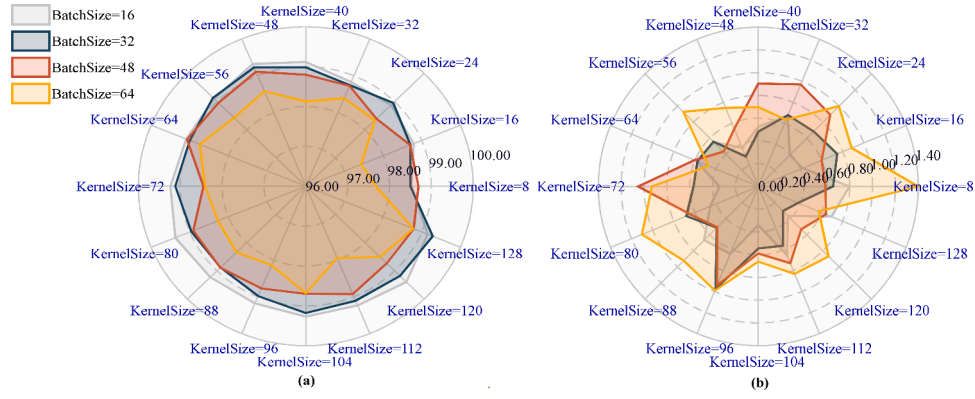


Fig. 9. Performances of (a) average recognition rates (%) and (b) standard deviation of recognition rates (%) with different parameters under 1200 r/min-50 % Load.

**Table 3**  
Recognition rates (%) under different parameters.

| Kernel Size | Batch Size 16 | Batch Size 32 | Batch Size 64 | Batch Size 128 |
|-------------|---------------|---------------|---------------|----------------|
| 8           | 98.53 ± 0.80  | 98.50 ± 0.65  | 98.69 ± 0.59  | 97.67 ± 1.40   |
| 16          | 98.79 ± 0.42  | 98.71 ± 0.75  | 98.70 ± 0.60  | 97.44 ± 0.88   |
| 24          | 98.88 ± 0.39  | 98.95 ± 0.68  | 98.39 ± 0.89  | 98.34 ± 1.00   |
| 32          | 98.83 ± 0.67  | 98.75 ± 0.68  | 98.73 ± 0.97  | 98.39 ± 0.63   |
| 40          | 99.12 ± 0.52  | 98.98 ± 0.48  | 98.80 ± 0.90  | 98.13 ± 0.70   |
| 48          | 99.32 ± 0.37  | 99.23 ± 0.29  | 99.11 ± 0.55  | 98.58 ± 0.75   |
| 56          | 99.10 ± 0.55  | 99.14 ± 0.56  | 98.95 ± 0.43  | 98.43 ± 0.93   |
| 64          | 99.09 ± 0.59  | 99.00 ± 0.58  | 99.07 ± 0.60  | 98.75 ± 0.47   |
| 72          | 99.24 ± 0.34  | 99.12 ± 0.57  | 98.45 ± 1.05  | 98.41 ± 0.93   |
| 80          | 99.37 ± 0.45  | 98.96 ± 0.68  | 98.91 ± 0.64  | 98.27 ± 1.10   |
| 88          | 99.23 ± 0.67  | 98.88 ± 0.51  | 98.89 ± 0.51  | 98.35 ± 0.92   |
| 96          | 99.18 ± 0.64  | 98.98 ± 0.98  | 98.77 ± 0.94  | 98.15 ± 0.99   |
| 104         | 99.27 ± 0.34  | 99.18 ± 0.55  | 98.70 ± 0.59  | 98.68 ± 0.66   |
| 112         | 99.24 ± 0.66  | 99.11 ± 0.56  | 98.93 ± 0.73  | 97.95 ± 0.83   |
| 120         | 99.40 ± 0.37  | 99.18 ± 0.31  | 98.74 ± 0.53  | 98.51 ± 0.87   |
| 128         | 99.15 ± 0.69  | 99.28 ± 0.38  | 98.79 ± 0.64  | 98.78 ± 0.57   |

based method) increased, and almost tended to be constant, and the accuracy rate averages of the WDCNN-based exhibited a larger fluctuation. Second, under the determined SNR, the average accuracy rate of the IDRSN-based method is almost larger than that of the DRSN-based method, which is also larger than the average accuracy rate of the WDCNN-based method. Third, with the increase in SNR, the accuracy rate standard deviations of the IDRSN-based method and the DRSN-based method decrease with a slight fluctuation, but the accuracy rate standard deviations of the WDCNN-based method have a larger fluctuation. Meanwhile, the accuracy rate standard deviations of the proposed IDRSN-based method, compared with that of the DRSN-based method and the WDCNN-based method, were the smallest.

In general, the above analysis indicates that the proposed diesel engine fault diagnosis method based on IDRSN outperforms the DRSN-based method and the WDCNN-based method in terms of the anti-noise performance and reliability of the training- and testing- models under the input signal with different SNR values.

#### 4.5. Performance on multiple loading conditions

To demonstrate the generalization and stability on multiple typical speed conditions (i.e.: 1000 r/min-50 % Load, 1200 r/min-50 % Load, 1500 r/min-50 % Load, 1800 r/min-50 % Load, and 1900 r/min-50 % Load), a comprehensive comparison is conducted among the three methods (i.e.: WDCNN, DRSN, and IDRSN). The results are illustrated in Fig. 12 and Table 6. Firstly, as the r/min increases, the average accuracy rates of the proposed IDRSN method are around 98 % with minimal fluctuations, which surpasses that of the DRSN-based method, and the WDCNN-based method with noticeable fluctuation. Secondly, at a determined r/min, the average accuracy rate of the IDRSN-based method outperforms that of the DRSN-based method, which is also larger than the average accuracy rate of the WDCNN-based method. Third, with the increase of r/min, the accuracy rate standard deviations of the IDRSN-based method and the DRSN-based method decrease with slight fluctuations, but the accuracy rate standard deviations of the WDCNN-based method have a larger fluctuation, while the accuracy rate standard deviations of the proposed method based on IDRSN, compared with that of the DRSN-based method and the WDCNN-based method, is the smallest.

In general, the above analysis indicates that, compared with the DRSN-based method and the WDCNN-based method, the proposed diesel engine fault diagnosis method based on IDRSN exhibits the best performance in terms of the anti-noise performance and reliability of the training- and testing- models under multiple speed conditions. Under

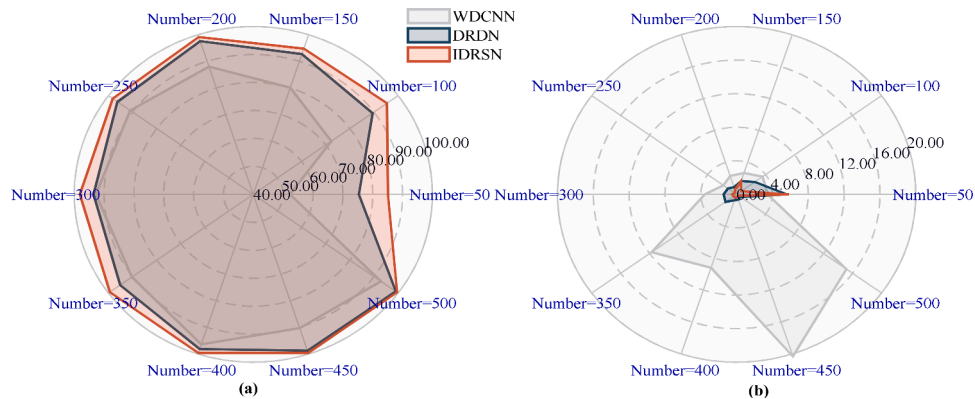


Fig. 10. Performances of (a) average recognition rates (%) and (b) standard deviation of recognition rates (%) under different sizes of training samples.

**Table 4**  
Recognition rates (%) under different sizes of training samples.

| Number of training samples / ×<br>11 | Methods       |              |                     |
|--------------------------------------|---------------|--------------|---------------------|
|                                      | WDCNN         | DRSN         | IDRSN               |
| 50                                   | 52.33 ± 3.46  | 75.37 ± 5.58 | <b>85.11 ± 5.71</b> |
| 100                                  | 72.36 ± 3.59  | 89.40 ± 2.52 | <b>95.39 ± 0.65</b> |
| 150                                  | 80.09 ± 2.69  | 92.71 ± 1.68 | <b>94.82 ± 1.62</b> |
| 200                                  | 88.02 ± 2.31  | 97.60 ± 0.91 | <b>99.18 ± 0.60</b> |
| 250                                  | 91.04 ± 1.83  | 96.26 ± 1.21 | <b>98.22 ± 0.27</b> |
| 300                                  | 92.74 ± 3.54  | 93.57 ± 1.45 | <b>98.11 ± 0.51</b> |
| 350                                  | 90.33 ± 11.72 | 95.16 ± 1.53 | <b>99.47 ± 0.42</b> |
| 400                                  | 96.42 ± 9.21  | 98.20 ± 0.73 | <b>99.60 ± 0.41</b> |
| 450                                  | 90.30 ± 20.30 | 98.79 ± 0.67 | <b>99.60 ± 0.32</b> |
| 500                                  | 92.88 ± 15.16 | 99.02 ± 0.52 | <b>99.57 ± 0.26</b> |

**Table 5**  
Recognition rates (%) under different SNRs.

| SNR / dB | Methods       | DRSN         | IDRSN               |
|----------|---------------|--------------|---------------------|
|          | WDCNN         |              |                     |
| −5.00    | 49.60 ± 12.94 | 74.65 ± 4.66 | <b>90.45 ± 1.48</b> |
| −4.00    | 66.93 ± 9.79  | 83.02 ± 3.54 | <b>92.88 ± 0.85</b> |
| −3.00    | 76.15 ± 8.10  | 86.90 ± 2.03 | <b>94.69 ± 1.41</b> |
| −2.00    | 84.96 ± 4.54  | 89.55 ± 3.01 | <b>96.30 ± 0.68</b> |
| −1.00    | 84.52 ± 7.12  | 90.83 ± 2.04 | <b>97.16 ± 0.69</b> |
| 0.00     | 86.03 ± 11.52 | 93.37 ± 1.43 | <b>97.66 ± 0.57</b> |
| 1.00     | 92.39 ± 5.82  | 92.39 ± 3.24 | <b>98.06 ± 0.51</b> |
| 2.00     | 90.36 ± 12.85 | 93.78 ± 1.51 | <b>98.60 ± 0.45</b> |
| 3.00     | 93.50 ± 10.08 | 95.25 ± 1.53 | <b>98.77 ± 0.36</b> |
| 4.00     | 95.43 ± 4.28  | 95.82 ± 1.11 | <b>98.98 ± 0.45</b> |
| 5.00     | 93.48 ± 12.05 | 96.12 ± 1.21 | <b>99.10 ± 0.30</b> |

multiple r/mins, the average accuracy rates of the IDRSN-based, DRSN-based, and WDCNN-based methods reach 98.38 % with a 0.42% error, 96.64 % with a 0.59 % error, and 93.56 % with a 5.07 % error, respectively.

#### 4.6. Visualization

To further illustrate the superiority of the proposed method, compared with the method based on DRSN and WDCNN models under

1200 r/min-50 % Load, Grad-CAM [32] was applied to visualize the class activation mapping (CAM), which means that the weight of the feature layers of the model, corresponding to the actual state (called label), are visualized with the gradient. The CAMs of the three methods above were calculated using the last feature layer (before global average pooling). As depicted in Fig. 13 the higher the color values, the higher the gradient values, and greater attention on the signal segments, which are beneficial for recognizing the state of the diesel engine. Fig. 13, along with Fig. 6, shows the following results:

Firstly, the CAMs of the above three methods under correct recognition indicate that the higher color segments are mainly located at EVS, IVS, PRS, and CS, and are also distributed in other shock segments, which may be caused by other working processes or other cylinder working cycles. Second, as shown in Fig. 13(a) and (b), the CAM of the proposed method based on IDRSN, compared with that of the based WDCNN method, indicates more concentrated and less disturbed activated signal segments with higher color values owing to the Residual Shrinkage Block Unit, which indicates the better recognition and anti-noise performance. Third, as shown in Fig. 13(a) and (c), the CAM of the proposed method, compared with that of the based DRSN method, indicates wider and less disturbed activated signal segments with higher color values owing to the wider kernel size, which means that the wider kernel size with a wider receptive field shows anti-interference performance.

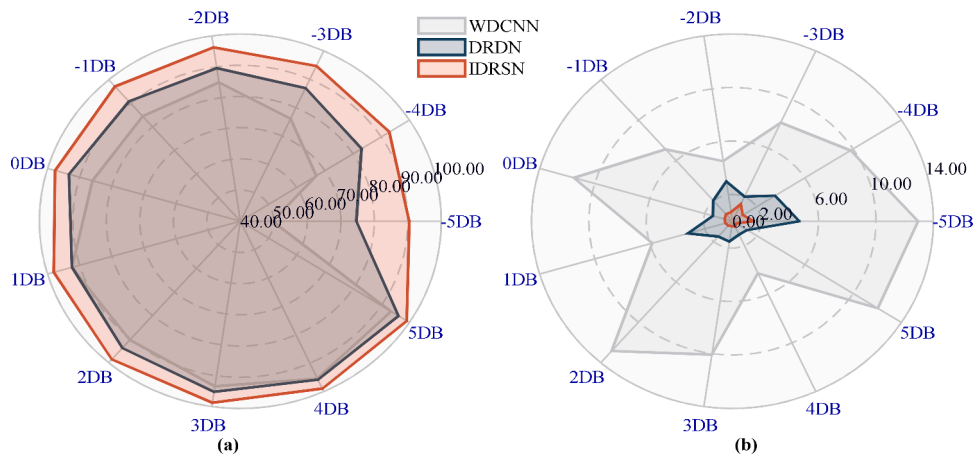
In conclusion, the above analysis indicates that, compared to the method based on WDCNN and DRSN, the proposed IDRSN-based method exhibits superior robustness, anti-interference, and anti-noise performance.

#### 4.7. Comparison of the existing literatures

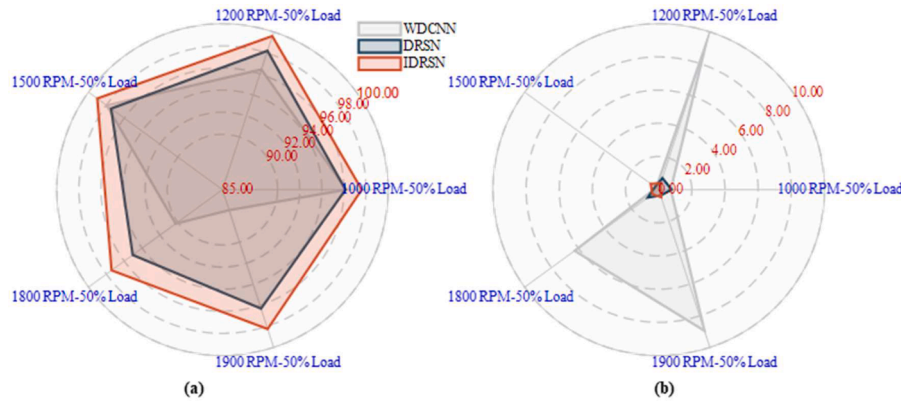
To further emphasize the superiority of the proposed IDRSN-based method, a comparison was conducted with the existing literatures. In Table 7, the methods, diesel engine states, and class number for classification and its accuracy in different and typical literatures are shown, which includes both the based-ML methods and the based-DL methods. The comparison results indicate the effectiveness and superior performance of the proposed method for diesel engine fault diagnosis.

### 5. Conclusions

In this paper, an improved deep residual shrinkage network (IDRSN) is developed to diagnose multiple faults at multiple degrees of a heavy-duty engine using single-channel surface vibration. The main conclusions, derived from comparisons with WDCNN and DRSN models on an engine test rig are summarized as follows:



**Fig. 11.** Performances of (a) average recognition rates (%) and (b) standard deviation of recognition rates (%) under different SNRs.



**Fig. 12.** Performances of (a) average recognition rates (%) and (b) standard deviation of recognition rates (%) under multiple speed conditions.

**Table 6**

Recognition rates (%) under multiple speed conditions.

| Conditions          | Methods             |                     |                     |
|---------------------|---------------------|---------------------|---------------------|
|                     | WDCNN               | DRSN                | IDRSN               |
| 1000r/min-50 % Load | 96.22 ± 0.73        | 96.15 ± 0.78        | <b>97.62 ± 0.48</b> |
| 1200r/min-50 % Load | 96.42 ± 9.21        | 98.20 ± 0.73        | <b>99.60 ± 0.47</b> |
| 1500r/min-50 % Load | 97.96 ± 0.19        | 97.47 ± 0.28        | <b>99.02 ± 0.60</b> |
| 1800r/min-50 % Load | 90.24 ± 6.27        | 95.05 ± 0.81        | <b>97.42 ± 0.47</b> |
| 1900r/min-50 % Load | 86.95 ± 8.95        | 96.31 ± 0.37        | <b>98.25 ± 0.48</b> |
| <b>Average</b>      | <b>93.56 ± 5.07</b> | <b>96.64 ± 0.59</b> | <b>98.38 ± 0.42</b> |

- (1) Analysis of recognition rate indicates that the IDRSN method exhibits better performance than the WDCNN and DRSN methods.
- (2) The effects of training sample number, signal-noise-ratio and multiple-working speeds for IDRSN model have been analyzed, which demonstrates the IDRSN's superior higher diagnostic accuracy, model stability and generalization of fault diagnosis under multiple-speed condntions.

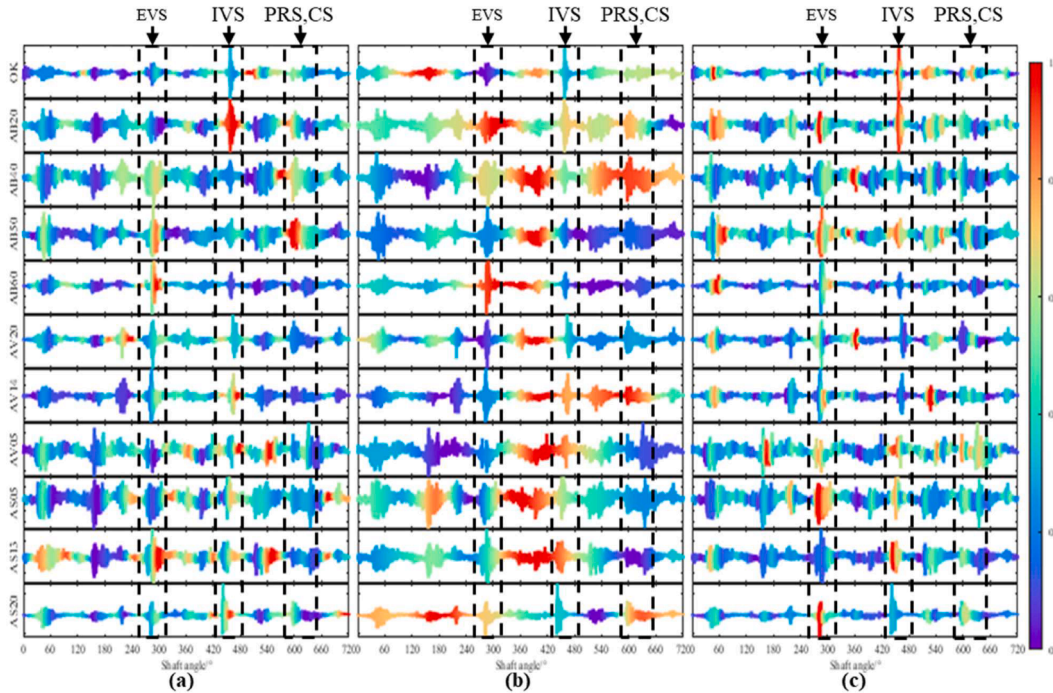
- (3) The visualization analysis with class activation maps indicates that the interpretations of IDRSN, DRSN, and WDCNN underscore the superior anti-interference capability of the proposed IDRSN-based method.
- (4) Future research directions include exploring channel attention mechanisms, transfer learning, and multi-source data fusion to address the unexplored correlations identified in this study.

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#### CRediT authorship contribution statement

**Xiaolong Zhu:** Writing – original draft, Visualization, Validation, Methodology, Investigation, Data curation. **Junhong Zhang:** Software, Resources, Project administration, Funding acquisition. **Xinwei Wang:** Software, Resources, Funding acquisition, Data curation. **Hui Wang:**



**Fig. 13.** Class activation maps of (a) IDRSN, (b) WDCNN and (c) DRSN.

**Table 7**  
Comparisons with existing literatures.

|      | Method              | Diesel engine states                                                                                        | Class number | Accuracy /% (Recognized samples / Total samples) |
|------|---------------------|-------------------------------------------------------------------------------------------------------------|--------------|--------------------------------------------------|
| [15] | WPD+NN              | Normal; Abnormal intake and Exhaust clearance                                                               | 9            | 91.8 % (413/450)                                 |
| [14] | VMD+RF              | Normal; Abnormal intake and exhaust clearance                                                               | 6            | 92.38% (388/420)                                 |
| [13] | MEMD+ DE            | Misfire                                                                                                     | 4            | 100 % (80/80)                                    |
| [18] | RCMFE+FASVM         | Normal; Abnormal intake clearance; Blocking air intake filter                                               | 10           | 98 % (490/500)                                   |
| [24] | CNN                 | Normal; Abnormal intake clearance                                                                           | 4            | 99.32 (–/–)                                      |
| [26] | RCNN                | Normal; Single Cylinder Misfire; Injection timing advance; Injection timing retard; Breakdown of air filter | 5            | 98.5 % (985/1000)                                |
| —    | The proposed method | Normal; Abnormal intake clearance; Blocking air intake filter                                               | 11           | 99.91 % (1099/1100)                              |

Software, Resources, Data curation. **Yedong Song:** Software, Resources, Funding acquisition. **Guobin Pei:** Visualization, Validation, Supervision, Investigation, Formal analysis, Data curation. **Xin Gou:** Visualization, Validation, Supervision, Investigation, Data curation. **Linlong Deng:** Visualization, Validation, Supervision, Formal analysis, Data curation. **Jiewei Lin:** Writing – review & editing, Software, Resources, Project administration, Funding acquisition, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

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